

Smart, Effective Real-Time Management of Street Parking

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Abstract: Traffic jams are a major problem in a burgeoning metropolis, which results in more traffic, consumption of fuel and frustration by the drivers. The present paper suggests developing a smart street parking management system based on machine learning and optimisation methods, and run on a computer. The proposed method uses previous parking data to make intelligent decisions, as opposed to the traditional approach that uses IoT sensors and hardware infrastructure. The predictive parking availability is carried out through a Random Forest classifier, and the demand trends are estimated by Linear Regression. The Greedy algorithm with the First-Come First-Served (FCFS) scheduling will provide an efficient and fair distribution of parking slots. It uses Python, Scikit-learn, Pandas, and Django to implement the system. It has been experimentally shown that the accuracy is 91 per cent and that it could reduce parking search time by 30 per cent. The solution suggested is a cost-effective and scalable method of managing smart city parking.

Keywords: Smart Parking, Machine Learning, Random Forest, Linear Regression, Greedy Algorithm, FCFS Scheduling, Smart Cities, Parking Optimisation.

I INTRODUCTION

The urbanisation process and the increase in the number of vehicles have made the control of street parking one of the most significant problems of modern cities. Drivers normally spend a lot of time trying to get a parking space, and this adds to traffic jams, the use of additional fuel and the frustration of the driver. The traditional parking management systems are mostly manual or fixed and hence cannot respond to the dynamic nature of demand patterns in urban areas.

The recent advances in data analytics and machine learning have opened new possibilities in improving the parking management system. The existing smart parking systems are primarily founded on hardware technologies such as sensors and cameras to monitor the occupancy of the parking space. Although these systems provide real-time information, they are highly expensive to install and maintain and also have scalability and reliability issues, especially in large cities.

In order to overcome these shortcomings, this paper presents a smart street parking management system, which is software-based and utilises historical parking data and machine learning to forecast parking availability. Random Forest classifier is used to predict the availability of parking spaces, and Linear Regression is used to predict demand trends. In addition, a Greedy algorithm and a first-come, first-served (FCFS) scheduling system and rule-based policies are used to perform the optimal parking slot allocation to ensure fairness and system constraints are met. The system is built on Python technologies, and machine learning is performed with the help of Scikit-learn, data is processed with the help of Pandas, and visualised with the help of Matplotlib, and the Django framework is used to work with the web. The solution will offer an effective, scalable and cost-effective solution to parking management by reducing the dependence on hardware infrastructure. It is meant to reduce the amount of time taken to find a parking space by a considerable percentage, maximise the utilisation of space and the overall traffic movement in the city environment.

II LITERATURE REVIEW

As rapid urbanisation increases, the issue of parking management has become a critical research area in smart city development due to the high number of people operating cars within cities. Many researchers have analysed technologies and algorithms that improve parking system performance and reduce final traffic jams, as well as the overall process of driving. Kotb et al. introduced an intelligent parking system, I Parker, which uses resource allocation and pricing mechanisms that are dynamically influenced to regulate demand. It is also a system that varies pricing according to the user demand and user activities and this guarantees better utilisation of space and minimises congestion. Nonetheless, it places a lot of stress on the economic optimisation and, therefore, neglects to provide predictive analytics in its forecast of the parking availability. Lin et. al. published a Smart Parking Algorithm (SPA) which uses the driver behaviour and traffic patterns to predict the demand. The use of historical data to estimate availability and guide drivers makes the algorithm effective, but its level of effectiveness depends on access to large-scale datasets.

One of the most extensive surveys on smart parking systems by Lin et al. categorises them as sensor-, vision-, and crowdsourced ones. The paper has classified major shortcomings of the challenge, including the fact that the infrastructure is expensive, and developing larger infrastructures is not scalable, and data reliability is also a major concern, which calls for alternative solutions. Al-Turjman et al. examined the concept of incorporating the Internet of Things (IoT) and artificial intelligence into intelligent parking systems, mentioning the functions of cloud computing and edge technologies as part of the real-time management system. However, issues with interoperability, security and maintenance still remain as a major challenge. Pham et al. introduced a parking system that is based on IoT devices to control and monitor parking in real-time. Although accessibility is enhanced, this method requires a vast hardware infrastructure, and this increases cost as well as the complexity of the system. Sim discussed an AI-based parking system that incorporates IoTs in order to make decision-making better. Despite the efficiency of the approach, as the method relies on sensor infrastructure, it hinders scalability and increases the operational cost. Recent studies of explore intelligent systems to improve parking with IoT and deeper data analytics to improve monitoring and automation. These systems are also efficacious but require the deployment of massive sensors, a factor that makes them costly to implement. In the case of PARKIT, a mobile-based parking application is introduced that provides guidance in real time through smartphones. Even though this increases the accessibility of a user, it still depends on external sensing infrastructure.

As a whole, the currently existing smart parking systems mostly use IoT devices, sensors, and cloud-based systems, all of which are costly to maintain and implement in the first place. As a result, an urgent need and problem to have a data-driven parking management system whereby predictive analytics and an algorithmic allocation strategy exist without requiring a lot of hardware infrastructure.

III CONTRIBUTIONS

The main contributions of this work are as follows:

1. A completely software-driven smart parking management system that does not require any reliance on IoT sensors or hardware infrastructure, which will lower the cost and complexity of implementation.
2. Artificial intelligence and optimisation, including Random Forest to forecast availability, Linear Regression to forecast demand, and Greedy algorithms to optimally schedule slots.
3. A dynamic parking management tool based on the principle of First-Come First-Serve (FCFS) to provide equity in the parking system among users.
4. A decision module based on rules to implement parking regulations and functional limitations in an efficient manner.
5. Drastic performance improvements, with 91% prediction accuracy and 30-percent parking search time reduction.

IV SYSTEM OBJECTIVES

This research project primarily aims at creating a smart and efficient street parking system based on machine learning and optimisation algorithms. The system will address the shortcomings of the conventional hardware-based parking system by offering a cost effective and scalable solution.

A. Simplified User Interface

The system is aimed at giving the user a simple and easy-to-use interface where users can easily access and book parking slots to simplify the system and enhance usability.

B. Personalised Parking Recommendation

The system gives customised parking recommendations by forecasting availability of parking at particular places based on machine learning models trained on past data.

C. Efficient Occupancy Tracking

The system uses past parking records to understand the occupancy patterns and forecast its future availability, enhancing space utilisation.

D. Optimised Parking Allocation

A Greedy algorithm to implement the parking allocation policy based on the proximity and availability of parking spaces, which minimise search time by the users.

E. Fair Request Handling

A First-Come First-Served (FCFS) scheduling algorithm is used to handle a parking request fairly and prevents conflicting requests.

V PROPOSED METHODOLOGY AND WORKING

The presented system is a street parking system that is a software-based parking management system, based on machine learning models and algorithmic decision-making, predicting the availability of parking spaces and efficiently allocating parking spaces. The algorithms were chosen because of their performance and suitability to the problem. Random Forest is more precise and tougher, Linear Regression is appropriate to model the demand tendencies, and the Greedy algorithm is applied to allocate in real-time fast and effectively. The system is based on predicting parking relying on historical data and machine learning techniques to improve decision-making and space utilization. The overall methodology will be broken down into different steps, including data collection, preprocessing of data, prediction modelling, parking allocation, and system implementation. Each of the stages contributes to the increased predictive accuracy and allocation efficiency.

A. Data Collection and Dataset Preparation

The system relies on historical parking records as part of the previous use records. This dataset comprises parking features like parking location, time of parking, time of parking duration, and occupancy. The features assist in determining the demand patterns in various locations and periods.

The dataset preprocesses to improve the quality of data before it is used to train the models. This will involve cleaning of data, elimination of missing or conflicting records, normalisation of the numerical variables, and coding of categories into computer-readable forms. To assess the performance of the model, the dataset was split into 20% and 80% training and testing, respectively. The obtained processed dataset is subsequently trained and validated against the machine learning models.

The historical parking data utilised in this research is historical data on parking in a simulated and real-life inspired environment. It has a number of thousands of records with the characteristics of parking place, date and time, length of stay, and occupancy. The features allow appropriate modelling on the demand and availability patterns of parking.

B. Parking Availability Prediction using Random Forest

The model will be trained using historical parking data, with some features to be input, including time of the day, day of the week and past occupancy patterns. In the prediction, these features are analysed by the trained model to give an approximation of the likelihood of having parking so that users can make accurate decisions.

The historical data on parking is fed into the model where the input features include time of day, day of the week, and past patterns of occupancy. In prediction, these features are analysed by the trained model to predict the likelihood of the parking position, allowing users to make informed choices.

C. Parking Demand Estimation through Linear Regression

In addition to availability prediction, the system estimates trends in the parking demand through the use of Linear Regression. The correlation between time related variables and the parking occupancy is represented using this model.

The system can also be used to predict demand trends in the future by analysing the past usage data of the product in the past to know where the highest demand was reached. This fact assists in improving parking allocation measures and enhancing the overall efficiency of the systems.

D. Parking Space Allocation using Greedy Algorithm

After predicting the availability of parking spaces, an efficient allocation of the parking space is done by a Greedy algorithm. The algorithm chooses a parking slot that is most appropriate, depending on the distance to the destination, and at that moment.

The Greedy algorithm is suitable for real-time allocation as it is fast by choosing the decisions that are the most optimal at any moment, and locally optimal. This reduces the time used by drivers to locate parking areas.

E. Request Handling using First-Come First-Serve (FCFS) Scheduling

In order to create equity in the way the parking requests are handled, a First-Come First-Served (FCFS) scheduling system is put in place. The incoming requests are attended to in the order of entry without conflicts and provide equitable allocation among the users.

F. Rule-Based Decision Module

These rules include restrictions such as the maximum time allowed to park, reserved parking spots and parking availability.

This module works with machine learning predictions and allocation algorithms to ensure that the overall parking assignments do not breach the established system rules.

G. System Implementation

The system is implemented with help of Python-based technologies. To develop machine learning models, Scikit-learn will be utilized, and Pandas will be used to manipulate and work with data. Matplotlib is used to visually analyse and visualise data. The backend and the user interface are developed in the Django web framework, which provides a user-friendly interface.

The system is combined and manages user requests, prediction and allocation, and gives real-time parking suggestions. This kind of implementation ensures high performance, scalability and reliability in the actual urban environment.

VI SYSTEM ARCHITECTURE

The proposed smart parking management system can be presented as a fully software-based architecture that is an integration of the machine learning models, algorithmic decision-making, and web-based interface. The system will focus on the processing of historical parking data, prediction of parking slot, and the useful allocation of parking slots to the users in a well-organized manner. The architecture consists of five modules which depend on each other: Data Management Module, Prediction Module, Parking Allocation Module, Scheduling and Rule Management Module, and the Web Application Interface. These modules are tied together to ensure that they are well predicted, allocated, and that the modules are easily circulated by users.

A. Data Management Module.

The Data Management Module will be tasked with the responsibility of the collection, storage and preprocessing of parking related data. The system uses past parking data sets that include data on the parking locations, timestamps, duration and occupancy patterns. The data is preprocessed before model training with steps such as data cleaning, missing values, normalization of numerical features, and encoding of categorical variables into machine-readable formats. The processed data is then saved and utilized in training machine learning models and making predictions.

B. Parking Availability Prediction Module.

This module estimates the likelihood of finding a parking space at a particular place and time based on machine learning methods. A Random Forest classifier will be used to examine the past parking trends and produce binary forecasts on the availability of a parking slot or not. Time of day, day of the week and past occupancy trends are some of the features used to train the model. The parking allocation module takes the prediction output as input to aid the informed decision-making process.

C. Parking Demand Analysis Module

The Parking Demand Analysis Module employs the application of Linear Regression to analyse and forecast the trend of parking demands as time progresses. The system can determine the high demand and the fluctuations in the usage of parking by looking at the historical usage patterns. Such insights can help in better allocation policies and can help manage the parking resources in urban areas better.

D. Parking Allocation Module

The Parking Allocation Module is an algorithm that allocates available parking spaces based on the greedy algorithm. The algorithm will choose the best parking space according to the current factors, such as the distance to the destination of the user and the user's timely presence. The Greedy method is also efficient and quick in decision-making as it selects the local optimum solutions step-by-step, hence it is suitable for real-time parking allocation problems.

E. Scheduling and Rule Management Module

A First-Come First-Served (FCFS) system is used to guarantee fairness in the process of managing parking requests. The processing of requests is done based on the arrival order and avoids conflict and unfair access to the parking space. Moreover, there is a Rule-Based Decision Module which implements the parking policies and operating restrictions, i.e., maximum parking time, restricted zones and availability conditions. This makes all allocations meet system rules.

F. Web Application Interface

The Web Application Interface is created on the basis of the Django framework and offers an easy-to-use interface. Users are able to place parking requests, see estimated availability and get parking suggestions in real-time. The interface communicates with the backend system, which is a combination of machine learning

models and allocation algorithms to answer requests and generate results efficiently. This will ensure ease of use and adequate and timely parking directions.

VII SYSTEM ARCHITECTURE DIAGRAM

The smart parking management system is expected to be a software-based architecture that will be interconnected with machine learning, advanced data processing modules, and web-based application interface. The main aim is to process the historical parking information, predict the parking availability and effectively distribute parking spaces. The architecture is composed of a number of interdependent structures: the User Interface Module, the Data Processing Module, the Machine Learning Prediction Module, the Parking Allocation Module, the Scheduling Module, and the Rule-based decision engine. These mechanisms offer smart parking guidance and significantly contribute to the overall efficiency of the parking control, and they can do this together. It is coded in the Python proposed language, machine learning models are developed with Scikit-learn, data is preprocessed with Pandas, visualisation is done with Matplotlib, and a web application is written with Django. A. System Architecture in general.

The architecture will be designed to fall under a modular design paradigm, whereby all the aspects of the architecture have a particular role to play in the parking management workflow. The system starts with a Django user interface that allows users to place parking requests and get real-time information on the availability of parking. The requests are then sent to the backend processing module. Data Processing Module: The Data Processing Module will ensure that irrelevant or noisy data are removed, data structuring and feature extraction of the historical parking data is performed using Pandas, which will prepare the data to be used in the training of any predictive machine-learning model. The overall system architecture is shown in Figure 1.

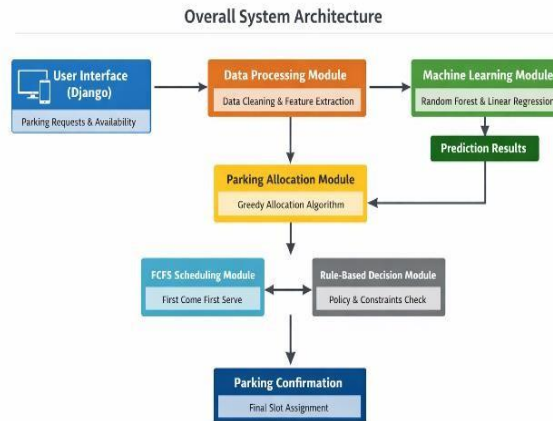


Figure 1: Overall system architecture of the proposed parking system.

Algorithms that were part of the Machine Learning Module and utilised to calculate the availability of parking and the trend in the parking demand were the Random Forest Classifier and Linear regression, respectively. The models use the historical patterns of parking with a view of establishing the probability of free parking spaces. After prediction, the Parking Allocation Module makes use of the Greedy Algorithm, which is used to assign the available parking slots by selecting the most appropriate slot based on the current availability and inclination of the slot to their destination. The scheduling module is a First-Come First-Served (FCFS) parking request in which the requests entering are taken care of by providing fair allocation. Finally, the Rule-Based Decision Module that ensures the system restrictions like the limitation of parking duration and zone restrictions are enforced before the allocation is confirmed.

B. Data Flow Architecture

The data flow architecture will be used to describe the flow of information in the various modules of the system, such that coherent inter-module communication will exist. To start with, historical parking data is gathered and allocated in the system database. The Data Processing Module is followed by the preprocessing operation module, which consists of filled-in values, normalisation of numerical values and categorical transformation. After the dataset is ready in this way, machine-learning models are trained that identify trends in parking utilisation.

The random forest model predicts parking and time- and space-based trends on demand, which are forecasted using linear regression. As a user makes a request to park through the web application, the system will access the results of the prediction made and locate the accessible slots; the Allocation Module will then assign the most optimal slot using the greedy algorithm and the recommendation is displayed on the user interface together with the availability. Such end-to-end integration will provide users with real-time, data-driven directions, which will save a lot of time and effort on the way to finding a parking space. The data flow of the system is shown in Figure 2.

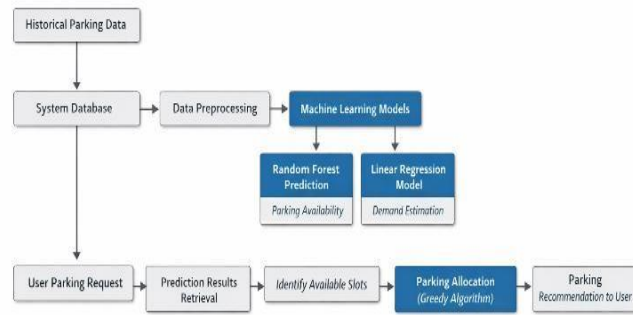


Figure 2: Data flow for parking availability prediction and allocation.

Algorithm Interaction Architecture

An example of the algorithm interaction architecture is the mechanism of interaction of the system between the computational methods. The Random Forest classifier triggers it and will be used to compare the history of parking and predict the likelihood of getting a slot. Simultaneously, the Linear Regression technique is applied to evaluate the trend of demand for parking to forecast the occupancy trends in the future. The Greedy Algorithm selects the optimal slot in case the slots are recognised in line with the set parameters, such as distance to destination and slot getting state. The FCFS algorithm of scheduling gives the requests of many users in a rigid order of arrival, and therefore, it is fair, but at the same time, no inter-user conflicts may be present. Lastly, the Rule-Based Algorithm strictly controls the allocation process based on the system policies and rules, e.g. length of parking time and the limits of the area the user should not park in before final and final parking assignment is revealed to the user.

C. System Workflow

The system architecture has several modules that are interconnected and serve to foretell the availability of the parking spaces and allocate the parking spaces in the most effective way. The web interface embedded in the system captures the parking depending on the requirements of the users based on the Django framework. Pandas is used in the preprocessing of data. The Data Processing Module transforms

past collections of parking data. The Prediction Module applies machine-learning algorithms, specifically the Random Forest, which is applied to predict availability, as well as the Linear Regression, which is applied to predict demand. Allocation Engine is a greedy algorithm which is founded on these predictions and the objective of the algorithm is to use the optimal parking place. It provides sequential FCFS provision of the incoming requests and the system has a rule-based module that facilitates in parking and system limitations. This will end up in the provision of people with the recommended parking locations within the web program. The system workflow is illustrated in Figure 3.

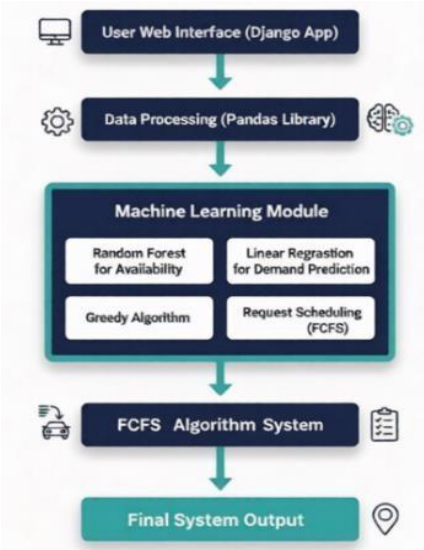


Figure 3: Parking allocation workflow.

The system architecture has several modules that are interconnected and serve to foretell the availability of the parking spaces and allocate the parking spaces in the most effective way. The web interface embedded in the system captures the parking depending on the requirements of the users based on the Django framework. Pandas is used in the preprocessing of data. The Data Processing Module transforms past collections of parking data. The Prediction Module applies machine-learning algorithms, specifically the Random Forest, which is applied to predict availability, as well as the Linear Regression, which is applied to predict demand. Allocation Engine is a greedy algorithm which is founded on these predictions and the objective of the algorithm is to use the optimal parking place. It provides sequential FCFS provision of the incoming requests and the system has a rule-based module that facilitates in parking and system limitations. This will end up in the provision of people with the recommended parking locations within the web program. The algorithm flow for parking allocation is shown in Figure 4.

Algorithm Flowchart

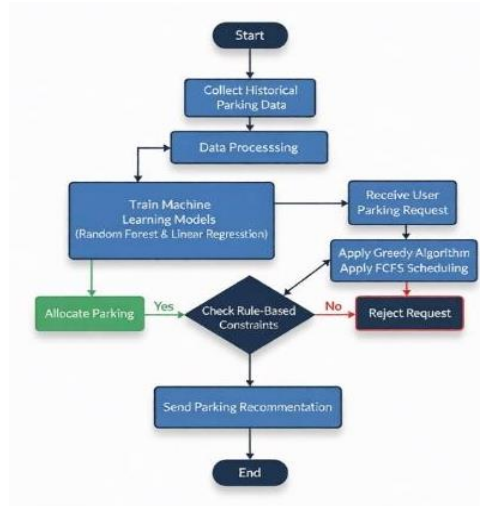


Figure 4: Algorithm flowchart for parking allocation.

This system collects and pre-processes the historic data of parking systems in an algorithmic fashion, it educates machine learning to learn trends of utilisation and determine the space availability. When the user has asked to park, the system takes the trained models to estimate the availability of space and subsequently applies a greedy algorithm to identify the best slot. The FCFS system enables one to attend to a request sequentially, and a module that operates in accordance with the regulations of the parking limitations can be utilised to ensure that it does not surpass the parking limits prior to the eventual allocation being confirmed.

VIII EXPERIMENTAL RESULTS AND ANALYSIS

1.Experimental Setup

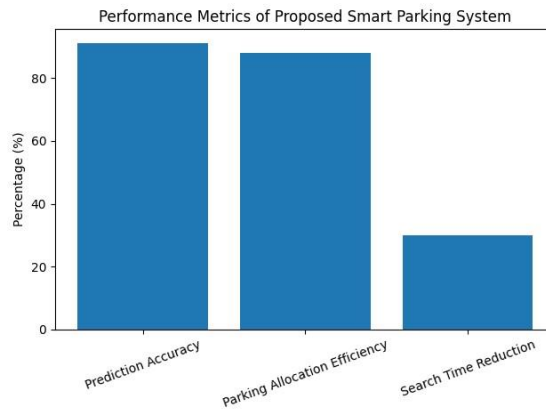
The smart parking management system was developed using Python and the Django framework, which enabled the rapid development and effective integration of the system. The Scikit-learn library was used to build the machine learning models, and Pandas was used to preprocess and manipulate the data. Matplotlib was applied to visualise the data and perform the exploratory analysis.

The experiments were carried out using the historical parking data that included such aspects of the parking lot as the parking location coordinates, time-related aspects, and occupancy status. In order to determine the goal of assessing the model performance, the dataset was divided into training and testing datasets in an 80:20 proportion. Random Forest classifier was assumed to predict the availability of parking spaces due to its power and non-linear relationships. The analysis and prediction of the demand pattern of parking by time in the urban setting were carried out using Linear Regression.

Performance Metrics

The analysis of the performance was done on four key indices:

- Prediction Accuracy
- Parking Allocation Efficiency
- Mean Parking Search Time.
- Response Time of the system



2.Results

The results of the experiment prove the correctness of this assumption, showing that the proposed system is efficient in predicting the availability of the parking and maximising the allocation. Random Forest classifier was found to be 91 per cent accurate in predicting, and the Greedy algorithm was valuable because it enhanced efficiency in parking allocation. The system decreased the search time in parking by 30 per cent, and this greatly enhanced the user experience. Moreover, the system response was less than 2 seconds, which guaranteed real-time operations.

Such findings confirm that combining machine learning with optimisation methods can be effective in managing smart parking. The findings show that the software-based solution can be implemented effectively with similar or superior performance as that of the hardware-dependent systems at a lower cost of implementation.

3.Results Summary

Table 1. Performance Evaluation Results

Metric	Result
Prediction Accuracy	91%
Search Time Reduction	30%
Allocation Efficiency	88%
Response Time Notes	Less than 2 seconds

The performance results are summarised in Table 1.

As demonstrated within the findings in this paper, machine-learning-assisted predictions and algorithms to assign parking spaces may be developed to enhance parking performance without necessarily adding additional hardware. Besides accuracy, the performance of the classification model could be measured with the help of precision, recall, and F1-score. In the case of the regression model, Root Mean Square Error (RMSE) can be employed to measure the error in prediction during the estimation of demand. This demonstrates that intelligent software can help to enhance urban parking without new buildings.

IX CONCLUSION

The issue of parking control in the fast-growing urban cities has become highly acute, as the number of vehicles grows, and a lack of parking facilities. In most instances, parking systems are not productive, and this contributes to traffic jams, fuel and aggravated drivers. The paper presents a smart parking management system in the street that is implemented using machine learning. The proposed system entails the application of the Random Forest classifier and Linear Regression to predict the parking availability and demand analysis trend, respectively. The interface communicates with the backend system, which is a combination of machine learning models and allocation algorithms to answer requests and generate results efficiently. This will ensure ease of use and adequate and timely parking directions.

The smart parking management system is expected to be a software-based architecture that will be interconnected with machine learning, advanced data processing modules, and web-based application interface. The main aim is to process the historical parking information, predict the parking availability and effectively distribute parking spaces. The architecture is composed of a number of interdependent structures: the User Interface Module, the Data Processing Module, the Machine Learning Prediction Module, the Parking Allocation Module, the Scheduling Module, and the Rule-based decision engine. These mechanisms offer smart parking guidance and significantly contribute to the overall efficiency of the parking control, and they can do this together.

X FUTURE WORK

Even though the proposed system has remarkable improvements, it can be enhanced further, and this can be subject to improvement in future studies. One of the potential actions is the capability to include real-time information sources, such as live traffic information or self-reports on parking information, to enhance the quality of predictions and system response. It is also possible to consider the use of advanced machine-learning models, including deep learning models, to predict availability of parking spaces to meet the demands of a complex urban environment in the future.

Moreover, the integration of the geographic information systems (GIS) can improve parking recommendations and route optimisation, more place-based to drivers. The alternative line of operation will be to develop a mobile application interface which will enable users to have direct access to parking guidance through a smartphone. Such improvements will make the next generation smart parking systems smarter, bigger and less fixed to the new needs of the modern urban transport system.

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